21 The functions f and g are such that

$$f(x) = 3x - 1 \quad \text{and} \quad g(x) = x^2 + 4$$

(a) Find $f^{-1}(x)$

(a) Let $y = 3x - 1$. Then $y + 1 = 3x$.

1 mark
Rearrange $y = 3x - 1$.

$x = (y + 1)/3$, so $f^{-1}(x) = (x + 1)/3$.

1 mark
$(x + 1)/3$.

$f^{-1}(x) =$ $(x + 1)/3$

(2)

Given that $fg(x) = 2gf(x)$,

(b) show that $15x^2 - 12x - 1 = 0$

(b) $fg(x) = f(x^2 + 4) = 3(x^2 + 4) - 1$

1 mark
Find fg(x).

$= 3x^2 + 11$

$gf(x) = g(3x - 1) = (3x - 1)^2 + 4$

1 mark
Find gf(x).

$= 9x^2 - 6x + 5$

1 mark
Expand $(3x - 1)^2$ correctly.

$fg(x) = 2gf(x)$ gives

1 mark
Equate fg(x) and 2gf(x).

$3x^2 + 11 = 2(9x^2 - 6x + 5)$

$= 18x^2 - 12x + 10$

Therefore $15x^2 - 12x - 1 = 0$.

1 mark
Reach the stated result from working.

(5)

(Total for Question 21 is 7 marks)

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